Aptitude Formula Sheet

Quick revision for placement tests

1. Percentages

- x% of y = y% of x.
- Increase by r% then decrease by r% -> net change = -(r^2/100)%.

2. Profit & Loss

- Profit% = (Profit / CP) * 100.
- SP = CP * (1 + Profit%/100) or CP * (1 - Loss%/100).
- Discount% = (Discount / MP) * 100.

3. Simple & Compound Interest

- SI = (P * R * T) / 100.
- CI = P * (1 + R/100)^T - P.

4. Time & Work

- Work = Rate * Time. Combined rate = sum of individual rates.
- If A is twice as good as B, A takes half the time.

5. Time, Speed & Distance

- Speed = Distance / Time.
- km/h to m/s: multiply by 5/18.
- Downstream = (boat + stream), Upstream = (boat - stream).

6. Averages

- Average = Sum of items / Number of items.
- Weighted average for groups.

7. Ratio & Proportion

- a:b = c:d implies ad = bc.
- Partnership: profit shared in ratio of (capital * time).

8. Probability & Combinations

- Probability = Favorable / Total.
- nCr = n! / (r! * (n-r)!), nPr = n! / (n-r)!.
- For 2 dice: total outcomes = 36.